Mahi Desai

+91 8779316909 | desaihmahi@gmail.com | linkedin.com/in/mahi-desai-0aa456380 | github.com/d-mahi14

Education

Veermata Jijabai Technological Institute, Matunga **Aug. 2024 - Present**
Third Year B.Tech in Computer Engineering. CGPA - **8.49**

Prakash College of Commerce and Science **Aug. 2022 - May. 2024**
MHT-CET | **99.95%ile** | HSC | **88.83%** | JEE Main | **98.84%ile**

Relevant Coursework

- Data Structures and Algorithms
- Linear Algebra
- Discrete Mathematics
- Artificial Intelligence
- Software Engineering
- Operating Systems
- Object Oriented Programming
- Database Management System
- Machine Learning

Projects

JobLink-AI-Powered Job Matching Platform | *React.js, Node.js* **Dec. 2025 - Feb. 2026**

- Built a full-stack job matching platform with AI-driven resume analysis using **Groq LLM**, extracting **5+ contact fields** and **6 skill categories**, storing structured data across **4 Supabase PostgreSQL tables** with RLS policies.
- Designed a RESTful API with **14+ endpoints** across **4 route modules** (auth, resumes, jobs, applications) featuring **JWT-based authentication** and role-based access control for **2 user types** (candidate/company).
- Implemented an AI match scoring pipeline cross-referencing parsed resume embeddings against job requirements, surfacing matched/missing skills with **85–99% confidence scores**, sorting N applicants by match score to reduce manual screening effort.

Assessing Biodiversity from eDNA Datasets | *Python, TensorFlow, HDBSCAN, NumPy* **Aug. 2025 - Nov. 2025**

- Engineered an AI-driven eDNA analysis pipeline processing **20,000+ DNA sequences**, compressing **256-dim k-mer vectors** into **32-dim VAE embeddings** — an **87.5% dimensionality reduction** — eliminating dependence on reference databases.
- Applied **HDBSCAN clustering** on VAE embeddings for de novo taxonomic identification, delineating **8+ known species** clusters and flagging **∼20% of clusters** as novel OTUs, with confidence scores of **85–99%**.
- Deployed a **Flask REST API** with end-to-end pipeline integration, serving **UMAP 2D visualizations** and **4 biodiversity indices** (Shannon, Simpson, Pielou, Richness) across **2 endpoints**.

Text-to-Handwriting | *Python, PyTorch* **Jul. 2025 - Oct. 2025**

- Designed and Engineered a **modular Conditional GAN (cGAN)** architecture with distinct **Content Encoder and Style Encoder** to achieve decoupled, fine-grained control over generated handwriting style and textual content.
- Trained a **PyTorch** model using adversarial, reconstruction and auxiliary objectives; final losses were: **Discriminator** ≈ 0.2328; **Generator** ≈ 12.3285; **Writer Classifier (style)** ≈ 0.0159; **Word Recogniser** ≈ 0.0132.
- Utilized the **IAM Word Dataset** derived from (**1539 pages** across **657 writers**) to synthesize open-vocabulary handwritten images, producing high-fidelity, user-specified handwriting styles across diverse writers.

Achievements

Neural Networks and Deep Learning **Nov. 2025**

- Completed Neural Networks and Deep Learning, covering architectures, forward/backward propagation, and optimization.

Improving Deep Neural Networks **Nov. 2025**

- Completed Improving Deep Neural Networks, covering regularization, dropout, optimization, tuning, and batch normalization.

Generative and Agentic AI **Apr. 2026**

- Completed Generative and Agentic AI, covering LLMs, RAG, AI agents, multimodal AI, and responsible AI.

AWS Cloud Practitioner Essentials **Sep. 2026**

- Completed AWS Training & Certification coursework covering core AWS services, cloud security, architecture, pricing, and cloud concepts.

Technical Skills

Languages: C++, Python, JavaScript, SQL, HTML/CSS
Developer Tools: Git, GitHub, VS Code, Google Colab, Kaggle, Jupyter Notebook, Postman
Frameworks/Libraries: React.js, Node.js, PyTorch, TensorFlow

Responsibilities

Core Member | *Project X, VJTI* **Jul. 2025 - Present**

- Member of **ProjectX**, a student-led technical club at VJTI focused on peer learning and mentorship.
- Contributed to open-source via **Hacktoberfest** and earned the **SuperContributor** badge.

Department Coordinator | *E-Cell, VJTI* **Sep. 2024 - Sep. 2025**

- Supported the **design team** in creating impactful visual content and event materials while fostering problem-solving, innovation, and collaboration within the team.